

Gujarat Police Smart AI CCTV System

Complete 6-Model Unified End-to-End Surveillance, Tracking & Investigation Suite

★ 6-Layer Architecture Quick Reference Matrix

#	AI Module / Engine	One-Line Question	Police Surveillance Pipeline Role
1	YOLO26 	KYA DIKH RAHA HAI?	Primary Visual AI Eye: Real-time person, car, bike, truck detection & bounding box coordinates.
2	ByteTrack / BoT-SORT 	KAHAN JA RAHA HAI?	Tracking & Movement Continuity: Assigns persistent Track IDs across video frames for trajectory mapping.
3	Plate Detector + PaddleOCR 	NUMBER KYA HAI?	Vehicle Identity Engine: Crops license plate & converts HSRP characters into searchable text.
4	FastReID / OSNet 	AUR KAHAN DIKHA?	Cross-Camera Person Search: Visual appearance embeddings for multi-camera suspect route search.
5	Qwen3-VL 	SCENE CONTEXT KYA HAI?	Contextual Reasoning Brain: Converts critical 10-sec alert video clips into natural-language summaries.
6	Vector Database 	AI DATA KAISE KHOJEIN?	Searchable AI Memory: Milvus/Qdrant vector index for sub-second similarity queries across the city.

1 YOLO26 — Primary Object Detection Model

Role: CCTV surveillance system ki primary "AI Aankh". Raw live camera stream se har single frame mein objects ko locate & classify karta hai.

Core Responsibility: Person (96%), Car (98%), Bike (89%) detection aur spatial coordinates (x1, y1, x2, y2) emit karna.

RTSP CCTV Stream → YOLO26 Inference Engine → Class + Bounding Box (x1, y1, x2, y2) → Downstream Trackers

2 ByteTrack / BoT-SORT — Object Tracking Engine

Role: Single-camera feed ke andar detect kiye gaye objects ko consistent **Tracking ID** (e.g. Person #17, Car #42) assign karta hai.

Why Needed: YOLO frame-to-frame association nahi karta. Tracker Kalman filter + IoU matching se temporal continuity aur loitering/speed alerts enable karta hai.

Frame t Detections → ByteTrack Association → Consistent Track ID (#17) → Spatial Route Trajectory

3 Plate Detector + PaddleOCR — ANPR Engine 🚗

Role: Vehicle registration plate isolate karke textual plate number (e.g. GJ01AB1234) extract karna.

2-Stage Architecture: Full frame par direct OCR chalane ke bajay, pehle *Plate Detector* plate crop karta hai, fir *PaddleOCR* HSRP characters read karta hai.

YOLO Vehicle Crop



Plate Localization



PaddleOCR (DBNet + SVTR)



RTO DB Sighting Timeline

4 FastReID / OSNet — Cross-Camera Person Search 🔍

Role: Jab koi suspect **CAM-01** se nikal kar **CAM-05** me jata hai, to visual appearance embeddings (512-dim) generate karke similarity match karna.

Guardrail: Yeh similarity/candidate matching hai (not definitive biometric identity proof). Final verification investigating officers karte hain.

CAM-01 Person Crop



FastReID Feature Vector



Vector Similarity Search



Multi-Camera Movement Map

5 Qwen3-VL — Video & Image Understanding Engine 🧠

Role: "AI Visual Brain" jo selected 10-sec alert video clips ka contextual reasoning karke **Natural Language Summary** generate karta hai.

```
// Qwen3-VL Context Alert
```

```
Trigger: Perimeter Fence Alarm | Summary: "Person in dark clothing breached north perimeter gate carrying a duffle bag."
```

6 Vector Database + Embeddings — AI Searchable Memory 📁

Role: Milvus / Qdrant cluster jo person appearance, vehicle descriptors, aur event vectors ko store karke sub-second search execute karta hai.

```
// Real-time Similarity Query
```

```
Query: Suspect Embedding Vector (512-d) → Top Candidate: CAM-04 (Pakwan Jn) @ 18:36 (94.2% Match)
```

Unified End-to-End Police Surveillance Architecture

CCTV
Ingestion



YOLO26
(Detection)



ByteTrack
(Trajectory)



PaddleOCR /
FastReID



Vector
DB



Command
Center

🗣️ 30-Second Hackathon & Official Presentation Pitch

"Gujarat Police AI CCTV project mein humne 6 modular layers implement kiye hain: **YOLO26** footage mein objects detect karta hai, **ByteTrack** unhe frame-to-frame track karta hai, aur **PaddleOCR** number plates read karta hai. Investigation ke liye **FastReID** suspects ko cross-camera search karta hai, **Qwen3-VL** incidents ka natural-language analysis karta hai, aur ye sabhi embeddings **Vector Database** me store hokar command center par instant AI search provide karte hain."